

GRCPC 2025 Editorial



A. zerorush

Author: Anastasis Koliopoulos.

Analysis: Anastasis Koliopoulos.

Tags: Greedy.

To begin, we reduce the problem to a decrease-only scenario by considering the absolute values of the sequence. Next, we observe that all operations must be applied to a single element — specifically, the smallest even number. This choice is optimal, since it will reach zero in the fewest steps among all even numbers. Note that an odd number can never become zero, so a sequence consisting entirely of odd numbers admits no solution. Let a denote the selected number, and let k be the number of operations needed for a to reach zero. Then

$$a - 2k = 0 \iff a = 2k \iff k = \frac{a}{2},$$

which is the answer to the problem.

The above approach requires only a single pass through the sequence, yielding a time complexity of $\mathcal{O}(N)$.

Indicative code

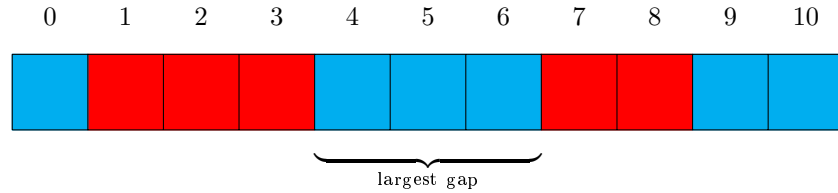
B. foodbreak

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Tags: Brute force, Sortings.

Let us first consider a simple, non-optimal but intuitive approach. We define an array $A[0 \dots T]$, where $A[i] = \text{red}$ if there is a presentation at timestamp i , and $A[i] = \text{blue}$ otherwise. For each presentation $i \in [n]$, we set every cell in the segment $[s_i, e_i)$ to red. After processing all presentations, we find the longest contiguous segment consisting of cells with blue color, which gives us the answer.



This algorithm runs in worst-case time $\mathcal{O}(N \cdot \max_{1 \leq i \leq n} \{e_i - s_i\} + T) = \mathcal{O}(N \cdot T + T) = \mathcal{O}(N \cdot T)$, and requires $\mathcal{O}(T)$ space, which is insufficient to meet the time and memory limits. Let us now consider a better approach.

C. mansion

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Analysis: Anastasis Koliopoulos.

Tags: Graphs, SSSP, Dijkstra's algorithm, 0/1 BFS.

We model the problem as follows. Let $G = (V, E)$ be a directed graph with $|V| = N$ and $|E| = M$, where each edge $e \in E$ has weight $w(e) \in \{0, 1\}$. The goal is to find the shortest path from vertex 1 to vertex N , or determine that no such path exists.

Since all edge weights are non-negative, we can apply *Dijkstra's algorithm* to compute the shortest path from vertex 1 to vertex N . Implementing Dijkstra's algorithm using an `std::set` or an `std::priority_queue` yields a time complexity of $\mathcal{O}((N + M) \cdot \log N)$, which is sufficient to pass all the test cases. Using a *Fibonacci heap* we can reduce the time complexity to $\mathcal{O}(N \cdot \log N + M)$, but its complex implementation and its large constant factor makes it impractical for competitive programming contests.

Actually, since all edge weights are either 0 or 1, we can instead use the more efficient *0/1 BFS* algorithm, which avoids the logarithmic factor and runs in $\mathcal{O}(N + M)$ time.

Indicative code

D. packages

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Tags: Binary search, k -th element.

We define the function $c : \{0, 1, \dots, n\} \rightarrow \mathbb{N}$, where $c(k)$ denotes the minimum cost to deliver k packages. We make the following crucial observation.

Observation. The cost function is non-decreasing.

Proof. We'll show that $c(k) \leq c(k+1)$, where $0 \leq k \leq n-1$. We denote by I_{k+1} the set of indices of the packages selected in an optimal solution with $k+1$ packages. For every $i \in I_{k+1}$ we have

$$\begin{aligned} k < k+1 &\implies k \cdot B[i] \leq (k+1) \cdot B[i] && (B[i] \geq 0) \\ &\implies A[i] + k \cdot B[i] \leq A[i] + (k+1) \cdot B[i]. \end{aligned}$$

Writing the above inequality for every i , we get

$$\left. \begin{array}{rcl} A[i_1] + k \cdot B[i_1] & \leq & A[i_1] + (k+1) \cdot B[i_1] \\ A[i_2] + k \cdot B[i_2] & \leq & A[i_2] + (k+1) \cdot B[i_2] \\ \vdots & & \\ A[i_{k+1}] + k \cdot B[i_{k+1}] & \leq & A[i_{k+1}] + (k+1) \cdot B[i_{k+1}] \end{array} \right\} \xRightarrow{+} \sum_{i \in I_{k+1}} A[i] + k \cdot B[i] \leq c(k+1).$$

Subtracting $A[j] + k \cdot B[j]$ from the left-hand side, for some $j \in I_{k+1}$, we obtain

$$\underbrace{\sum_{i \in I_{k+1} \setminus \{j\}} A[i] + k \cdot B[i]}_{\text{a feasible } k\text{-packages solution}} \geq c(k).$$

Combining the two inequalities above, we get

$$c(k) \leq c(k+1),$$

which completes the proof. ■

Leveraging the monotonicity of the cost function, we can *binary search the answer*. We choose a value $k \in \{0, 1, \dots, n\}$ and compute $c(k)$. If $c(k) \leq X$, where X denotes the available budget, we continue searching among values $k' \geq k$; otherwise, we restrict the search to values $k' < k$.

To determine which packages to deliver for a fixed k , we create the array $C[1 \dots n]$, where

$$C[i] = A[i] + k \cdot B[i]$$

and retrieve the k cheapest packages. There are three ways to obtain the desired packages, described below:

- Sort the array C in $\mathcal{O}(N \cdot \log N)$ time.
- Use the built-in C++ function `nth_element`, which runs in $\mathcal{O}(N)$ time on average.
- Use the *Median of medians algorithm*, which guarantees $\mathcal{O}(N)$ time in the worst case.

The binary search contributes an additional $\mathcal{O}(\log N)$ factor.

Overall, the best achievable time complexity is $\mathcal{O}(N \cdot \log N)$ in the worst case. Nevertheless, all of the above approaches are efficient enough to pass the given constraints. In the provided implementation, we use the second approach.

Indicative code

E. javelins

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Tags: Exchange arguments, DP, State rotation, Space optimization.

We will apply *Dynamic Programming (DP)* to solve this problem.

Before discussing the DP solution, observe that the order in which the targets are hit matters. Therefore, it is not sufficient to decide only which subset of targets to select, but also in which order to hit them.

Let I , where $I \subseteq [n]$ and $|I| \geq 2$, denote the set of the selected targets. Suppose we select two targets $i, i' \in I$. When is it optimal to hit target i before target i' ?

Let j, j' , with $1 \leq j < j' \leq |I|$, denote two time slots corresponding to the order in which the selected targets will be hit.

If we first assign target i to slot j and then target i' to slot j' , the energy consumption is given by

$$E = A[i] + (j - 1) \cdot B[i] + A[i'] + (j' - 1) \cdot B[i'].$$

If we swap the order in which we hit these two targets, the energy consumption is equal to

$$E' = A[i] + (j' - 1) \cdot B[i] + A[i'] + (j - 1) \cdot B[i'].$$

Solving the inequality below, we get

$$\begin{aligned} E < E' &\iff (j - 1) \cdot B[i] + (j' - 1) \cdot B[i'] < (j' - 1) \cdot B[i] + (j - 1) \cdot B[i'] \\ &\iff (j - 1) \cdot B[i] - (j' - 1) \cdot B[i] < (j - 1) \cdot B[i'] - (j' - 1) \cdot B[i'] \\ &\iff (j - j') \cdot B[i] < (j - j') \cdot B[i'] \\ &\iff B[i] > B[i']. \end{aligned}$$

This means that we hit target i before i' if and only if $B[i] > B[i']$.

Let's now determine the optimal set of targets using DP. The five stages of dynamic programming are specified as follows.

- (Definition of the DP table). We define $dp[i][j]$, where $0 \leq i, j \leq N$, as the minimum energy required to hit j out of the first i targets.
- (Choices). At each step i , we decide whether to skip target i or to include it as the j -th hit target.

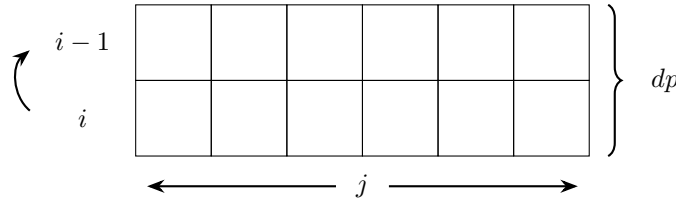
- (Transitions). The following recurrence is derived from the above choices:

$$dp[i][j] = \begin{cases} 0, & i = 0 \text{ and } j = 0, \\ \infty, & i = 0 \text{ and } 1 \leq j \leq N, \\ dp[i-1][0], & 1 \leq i \leq N \text{ and } j = 0, \\ \min \begin{cases} dp[i-1][j], \\ dp[i-1][j-1] + A[i] + B[i] \cdot (j-1), \end{cases} & 1 \leq i \leq N \text{ and } 1 \leq j \leq N. \end{cases}$$

- (Order of computation). The values are computed in increasing order of the index i .
- (Original problem solution). The solution is

$$\max_{0 \leq j \leq N} \{j : dp[N][j] \leq E\}.$$

The final observation is that, in order to compute any DP value in row i , we only need to refer to row $i-1$. Therefore, at any step, instead of storing the entire DP table in memory, we can keep only the current and the previous row.



The time complexity of computing this DP recurrence relation is $\mathcal{O}(N^2)$ and the space complexity is $\mathcal{O}(N)$.

Indicative code

F. audit

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Analysis: Anastasis Koliopoulos.

Tags: Offline queries, Sparse segment tree, Ordered set.

$$dp[i][j] = \begin{cases} 0, & i = 0 \text{ and } j = 0 \\ \infty, & i = 0 \text{ and } 1 \leq j \leq N, \\ dp[i-1][0], & 1 \leq i \leq N \text{ and } j = 0, \\ \min \begin{cases} dp[i-1][j], \\ dp[i-1][j-1] + A[i] + B[i] \cdot (j-1), \end{cases} & 1 \leq i \leq N \text{ and } 1 \leq j \leq N. \end{cases}$$